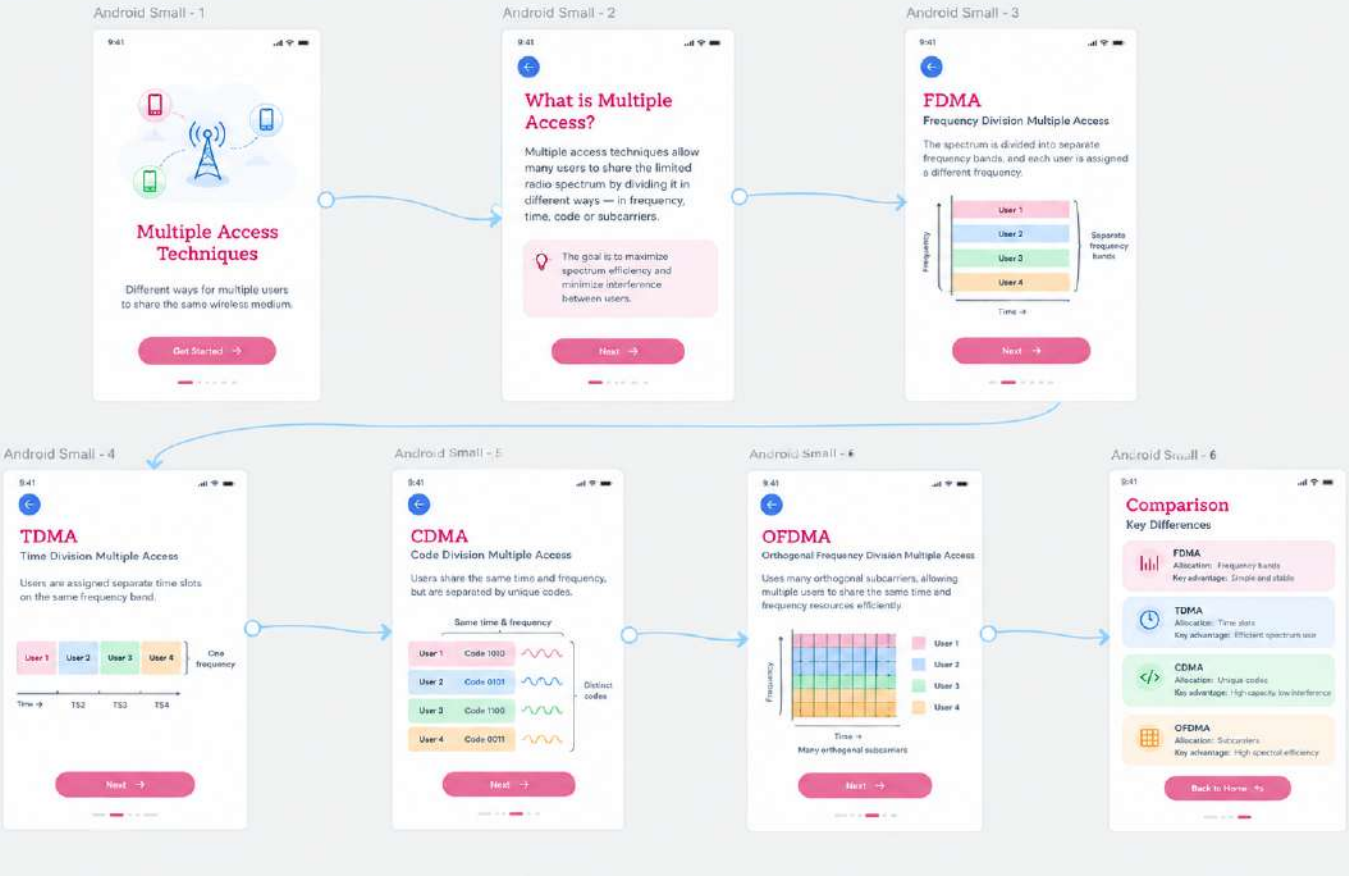
Android Small - 1
Multiple Access Techniques
Different ways for multiple users to share the same wireless medium.
Get Started
Android Small - 2
What is Multiple Access?
Multiple access techniques allow many users to share the limited radio spectrum by dividing it in different ways — in frequency, time, code or subcarriers.
The goal is to maximize spectrum efficiency and minimize interference between users.
Next
Android Small - 3
FDMA
Frequency Division Multiple Access
The spectrum is divided into separate frequency bands, and each user is assigned a different frequency.
User 1
User 2
User 3
User 4
Frequency
Separate frequency bands
Time
Next
Android Small - 4
TDMA
Time Division Multiple Access
Users are assigned separate time slots on the same frequency band.
User 1
User 2
User 3
User 4
One frequency
Time
TS2
TS3
TS4
Next
Android Small - 5
CDMA
Code Division Multiple Access
Users share the same time and frequency, but are separated by unique codes.
Same time & frequency
User 1
Code 1010
User 2
Code 0101
User 3
Code 1100
User 4
Code 0011
Distinct codes
Next
Android Small - 6
OFDMA
Orthogonal Frequency Division Multiple Access
Uses many orthogonal subcarriers, allowing multiple users to share the same time and frequency resources efficiently
Frequency
Time
Many orthogonal subcarriers
User 1
User 2
User 3
User 4
Next
Android Small - 6
Comparison
Key Differences
FDMA
Allocation: Frequency bands
Key advantage: Simple and stable
TDMA
Allocation: Time slots
Key advantage: Efficient spectrum use
CDMA
Allocation: Unique codes
Key advantage: High capacity, low interference
OFDMA
Allocation: Subcarriers
Key advantage: High spectral efficiency
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